

2D Beam Stress & Deflection Simulator Pro

Complete Project Documentation — Basic to Advanced

Platform:	Python 3.14 + Streamlit Cloud
GitHub:	github.com/rupanshujangid-collab/Beam-Simulator
Live App:	beam-simulator.streamlit.app
Version:	Pro — All Features Enabled
Features:	SFD · BMD · 3D · Dynamic · Excel · PDF · AI

[SFD/BMD](#)[Stress Heatmap](#)[3D View](#)[Dynamic Loads](#)[Excel Export](#)[PDF Reports](#)[10 Materials](#)[AI Suggestions](#)

[GitHub Repository](#)

[Live App — beam-simulator.streamlit.app](#)

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01 Project Overview & Purpose

The **2D Beam Stress & Deflection Simulator Pro** is a full-featured structural engineering web application built using Python and Streamlit. It enables engineers, students, and researchers to perform complete beam analysis — computing Shear Force, Bending Moment, Deflection, Bending Stress, and Factor of Safety — entirely in real time through an interactive browser interface.

The project bridges the gap between expensive FEA software like ANSYS/ABAQUS and simple hand calculations — providing ANSYS-style stress heatmaps, 3D beam visualization, moving load influence lines, material comparison, and automated AI engineering suggestions — completely free and deployed on Streamlit Cloud.

Key Goals

- Accurate beam analysis using classical Euler-Bernoulli beam theory
- 3 beam types: Simply Supported, Cantilever, Fixed-Fixed
- 4 cross sections: Rectangular, Circular, I-Beam, T-Beam
- 10 engineering materials with real properties
- Visual outputs: SFD, BMD, Deflection, Stress Heatmap, 3D View
- Moving load / dynamic analysis with influence lines and envelopes
- Professional reporting: Excel (3 sheets with charts) + PDF
- Deployable on Streamlit Cloud — accessible from any device
- Mobile-friendly with cross-platform zoom controls
- AI-powered engineering suggestions and safety assessment

[GitHub: github.com/rupanshujangid-collab/Beam-Simulator](https://github.com/rupanshujangid-collab/Beam-Simulator)

[Live App: beam-simulator.streamlit.app](https://beam-simulator.streamlit.app)

02 Tech Stack & Libraries

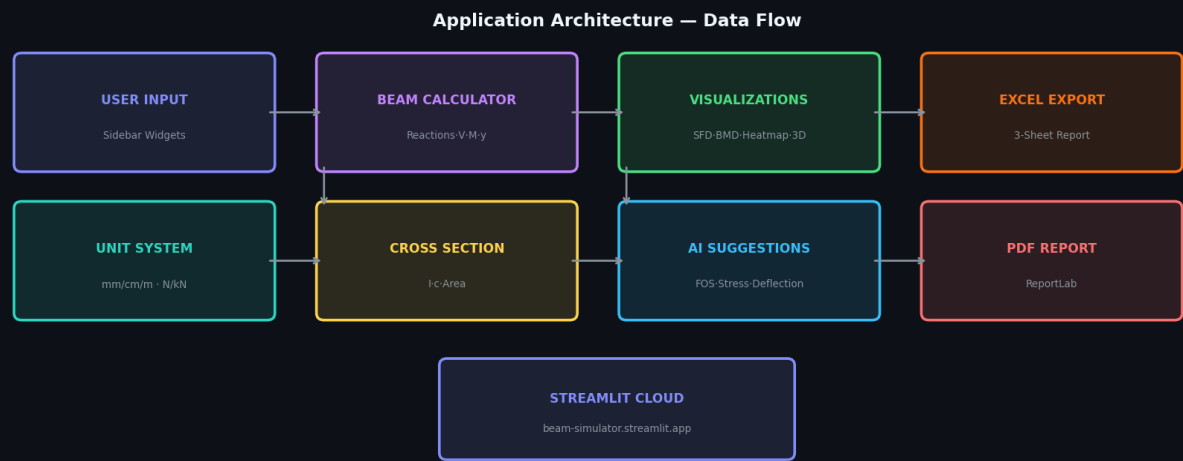
Library	Version	Role	Used For
streamlit	1.58.0	Web Framework	UI, sidebar, widgets, layout, download buttons
numpy	Latest	Numerics	Arrays, linspace, interp, where, math ops
matplotlib	Latest	Plotting	SFD, BMD, Deflection, Heatmap, 3D views
scipy	Latest	Science	Scientific computing support
reportlab	Latest	PDF Engine	Professional PDF report generation
openpyxl	Latest	Excel	3-sheet Excel with charts and styling
pandas	Latest	DataFrames	Material comparison table display
mpl_toolkits	Bundled	3D Plot	3D beam visualization

requirements.txt

```
streamlit matplotlib numpy scipy reportlab openpyxl pandas
```

03 Application Architecture

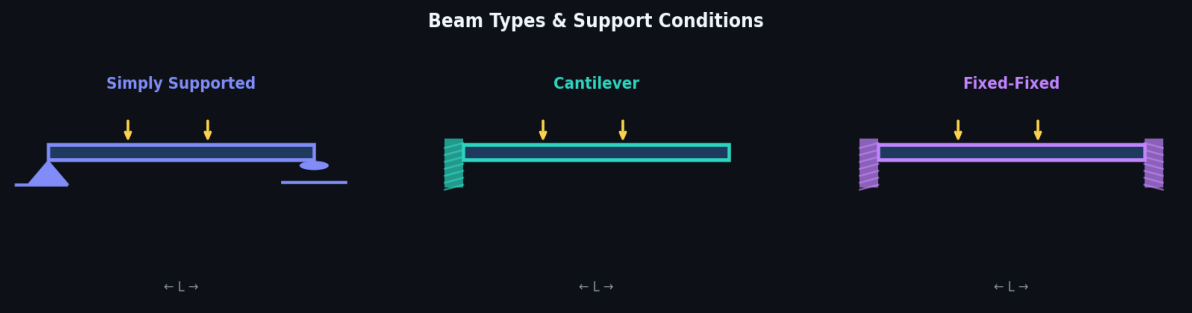
The application follows a linear single-file architecture — all logic, UI, calculations, and exports contained in `beam_simulator.py`. Data flows from user input through calculation engine to visualization and export modules.



File Structure

File	Size	Purpose
beam_simulator.py	56.5 KB / 1280 lines	Main app — all logic, UI, calculations, exports
requirements.txt	< 1 KB	Python dependencies for Streamlit Cloud deployment
README.md	2 KB	GitHub project description and usage guide
.gitignore	< 1 KB	Git ignore rules for pycache, .env, etc.
LICENSE	< 1 KB	MIT License for open source usage

04 Beam Types & Support Conditions



Simply Supported

Description	Pin at A, Roller at B — 2 unknown reactions. Statically determinate.
Equations	$RA = \Sigma[P \cdot (L - a) / L] + wL / 2 \mid RB = \text{Total Load} - RA$
Applications	Floor beams, bridge girders, simple spans

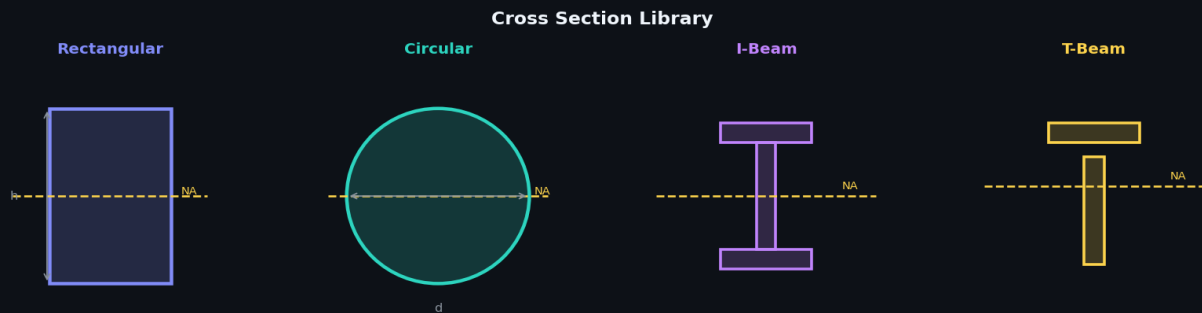
Cantilever

Description	Fixed at A, Free at B — provides reaction force RA and fixed moment MA.
Equations	$RA = \Sigma P + wL \mid MA = \Sigma[P \cdot (L - a)] + wL^2 / 2$
Applications	Balconies, diving boards, overhanging roofs

Fixed-Fixed

Description	Fixed at both ends — statically indeterminate. Reduces max moment by ~50%.
Equations	$RA = \Sigma[P \cdot (L - a)^2 \cdot (2a + L) / L^3] + wL / 2 \mid MA = \Sigma[P \cdot a \cdot (L - a)^2 / L^2] + wL^2 / 12$
Applications	Building frames, continuous beams, rigid connections

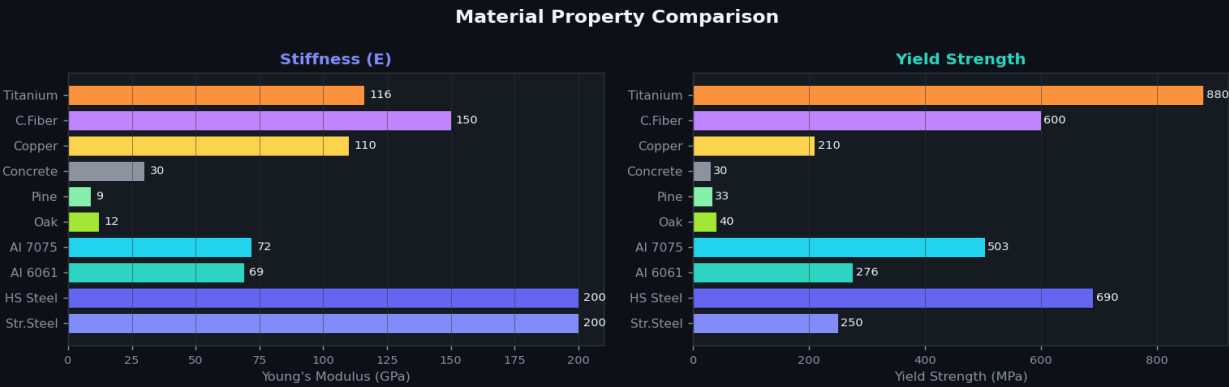
05 Cross Section Library



Section	Formula for I	c (to NA)	Best For
Rectangular	$I = b \cdot h^3 / 12$	$h/2$	General purpose, timber, concrete beams
Circular	$I = \pi \cdot d^4 / 64$	$d/2$	Shafts, pipes, rods, cylindrical members
I-Beam	$I = b_f \cdot h_t^3 / 12 - (b_f - t_w) \cdot h_w^3 / 12$	$h_t/2$	Steel structures, bridges, columns
T-Beam	Parallel axis theorem on flange+web	$\max(\bar{y}_f, h_t - \bar{y}_f)$	Concrete slabs, composite sections

The Neutral Axis (NA) is the axis about which bending stress is zero. For symmetric sections (Rectangular, Circular, I-Beam) it passes through the centroid at mid-height. For T-Beam, centroid $\bar{y}$ is computed using the parallel axis theorem: $\bar{y} = [A_f \cdot (h_t - t_f/2) + A_w \cdot (h_w/2)] / (A_f + A_w)$

06 Material Library — 10 Materials



Material	E (GPa)	Yield (MPa)	Density (kg/m³)	Key Notes
Structural Steel	200	250	7850	Most common structural material worldwide
High-Strength Steel	200	690	7850	Bridges, high-load structures, cranes
Aluminum 6061	69	276	2700	Lightweight, corrosion resistant, general use
Aluminum 7075	72	503	2810	Aerospace — highest strength Al alloy
Oak Wood	12	40	700	Traditional construction, furniture
Pine Wood	9	33	550	Light framing, low-cost construction
Concrete	30	30	2400	Weak in tension — always needs rebar!
Copper	110	210	8900	Electrical components, heat exchangers
Carbon Fiber	150	600	1600	Aerospace, F1, ultra-high performance
Titanium	116	880	4500	Medical implants, aerospace, marine

07 Load Types

Point Load (Concentrated)

A concentrated force P applied at position a from left support. Up to 8 point loads are supported simultaneously. Each load causes a step discontinuity in SFD and a kink (change in slope) in BMD. Position and magnitude are independently adjustable.

Applications: Buildings, bridges, crane loads, machinery mounts

UDL — Uniformly Distributed Load

Load of intensity w (N/m) spread uniformly over entire beam length. Produces linear variation in SFD and parabolic BMD. Equivalent point load = $w \cdot L$ acting at beam center. Contributions: $R_A = wL/2$, $R_B = wL/2$ for simply supported.

Applications: Self-weight, floor loads, snow loads, wind pressure

Dynamic / Moving Load

A moving point load P traverses the beam at 20 discrete positions from $x=0.01$ to $x=L-0.01$. At each position, full beam analysis is computed. Outputs include: BMD envelope (max/min), Deflection envelope (worst case), and Influence line for midspan moment. Critical for bridge girder and crane runway design.

Applications: Bridge design, crane runways, vehicle loading

08 Engineering Diagrams — SFD, BMD, Deflection

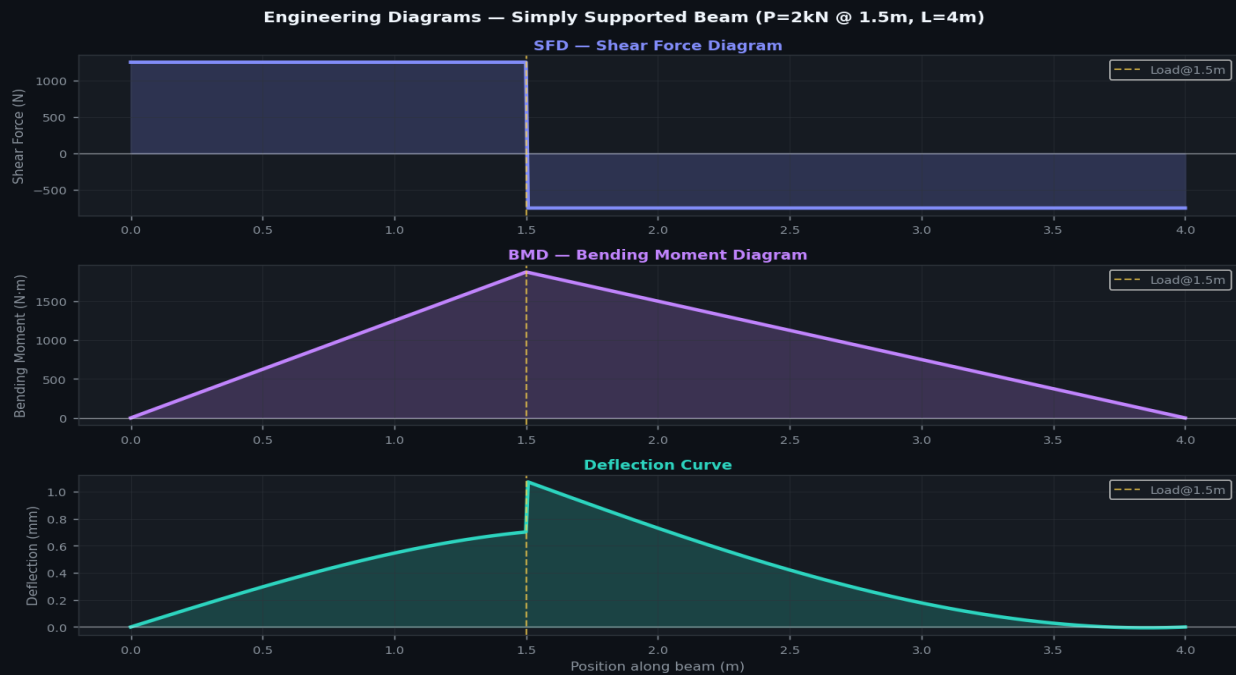
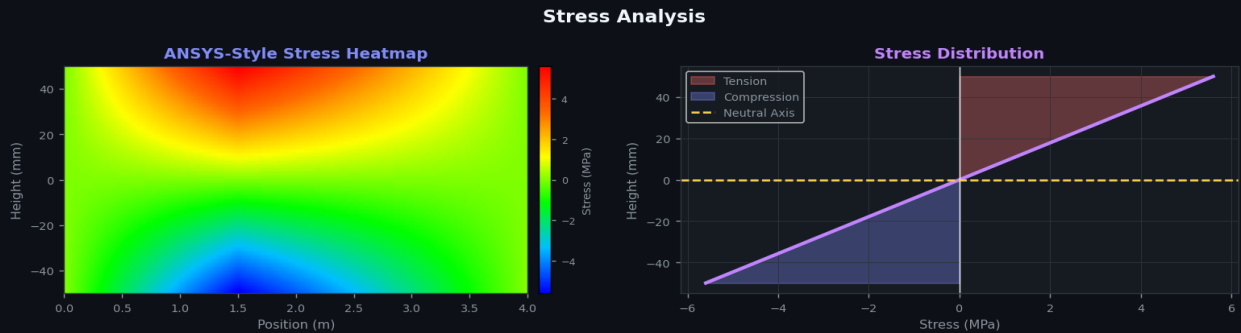


Diagram	What It Shows	Key Formula
SFD	Shear Force $V(x)$ — internal force perpendicular to beam axis	$V = RA - \sum[P \cdot H(x-a)] - w \cdot x$
BMD	Bending Moment $M(x)$ — internal moment causing beam to bend	$M = RA \cdot x - w \cdot x^2/2 - \sum[P \cdot \max(x-a, 0)]$
Deflection	Transverse displacement $y(x)$ — beam shape under load (mm)	$y = Pbx/(6EIL) \cdot (L^2 - b^2 - x^2)$ for x less than a [SS beam]

All three diagrams use superposition — contributions from each load are computed separately and summed. 2000 points computed along beam for smooth curves. Deflection displayed in mm (internal calculation in metres).

09 Stress Analysis & Heatmap



Bending stress at any point (x, y) : $\sigma(x, y) = M(x) \cdot y / I$. A 300×80 grid is computed — x along beam length (300 pts), y across section height (80 pts). Positive y (above NA) = Compression for positive M . Negative y (below NA) = Tension for positive M . Displayed as ANSYS-style colormap: Blue=max compression, Red=max tension.

$\sigma_{max} = M_{max} \cdot c / I / 1e6$ [MPa] where c = distance from NA to extreme fiber.

10 3D Visualization

Uses matplotlib's `mpl_toolkits.mplot3d`. Left subplot renders the beam as two surface patches (top and bottom faces) colored by material, with the deflected neutral axis shown as a red line. Load arrows rendered using `ax.quiver()`. Right subplot shows stress + deflection overlay in 2D side view with deflection curve scaled $\times 5$ for visibility.

11 Dynamic / Moving Load Analysis

Moving load P traverses 20 positions from $x=0.01$ to $x=L-0.01$. At each position, complete beam analysis is performed for the selected beam type. Three output plots:

Output	Description	Use
BMD Envelope	Max/min moment at every point for any load position	Design for worst-case moment
Deflection Envelope	Maximum deflection at each point across all positions	Serviceability check
Influence Line	Midspan moment vs load position	Identify critical load position

12 Safety Checker & Factor of Safety

Factor of Safety — Safety Zones



$FOS = \text{Yield Strength (MPa)} / \sigma_{\text{max}}$. Beam weight = $\rho \cdot A_{\text{cross}} \cdot L \cdot g$ [N].

FOS Range	Status	Meaning	Action Required
< 1.0	CRITICAL	Immediate failure expected	Complete redesign — change section/material
1.0 – 1.5	UNSAFE	Below minimum safe limit	Increase section size or use stronger material
1.5 – 2.0	MARGINAL	Below recommended 2.0	Consider increasing FOS to 2.0 minimum
2.0 – 3.0	SAFE	Meets standard requirements	Acceptable for most structural applications
> 3.0	OVER-DESIGNED	Excess material/weight	Consider lighter material or smaller section

13 AI Engineering Suggestions

The `ai_suggestions()` function analyzes all computed results and produces categorized recommendations: Warnings (red), Suggestions (white), Good news (green). 16 intelligent checks are performed:

FOS Checks

- $FOS < 1.0 \rightarrow$ Critical failure
- $FOS < 1.5 \rightarrow$ Dangerously low
- $FOS < 2.0 \rightarrow$ Below recommended
- $FOS \geq 2.0 \rightarrow$ Safe confirmation

Stress Checks

- $Stress > 90\% \text{ yield} \rightarrow$ Red alert
- $Stress > 60\% \text{ yield} \rightarrow$ Yellow warning
- $Stress < 60\% \text{ yield} \rightarrow$ Safe confirmation

Deflection

- $y_{\text{max}} > L/200 \rightarrow$ Serviceability failure + fix suggestion
- $y_{\text{max}} \leq L/200 \rightarrow$ Deflection within limit

Section Advice

- Rectangular: new height formula for $FOS \geq 2.0$
- Circular: new diameter formula for $FOS \geq 2.0$
- I-Beam: confirms efficient choice
- T-Beam: confirms uni-directional suitability

Material Advice

- Steel + $FOS > 10 \rightarrow$ Suggest aluminum (65% lighter)
- Aluminum + low FOS $\rightarrow$ Suggest steel
- Concrete $\rightarrow$ Warn about tension weakness
- Carbon fiber $\rightarrow$ Note high performance

Beam Type

- Simply Supported + low FOS $\rightarrow$ Suggest Fixed-Fixed
- Cantilever + low FOS $\rightarrow$ Suggest intermediate support

14 Excel Export — 3 Sheet Report

Sheet	Contents	Features
1 — Summary	All inputs + results	Color-coded cells, SAFE/UNSAFE in green/red, styled headers
2 — SFD_BMD_Data	Raw x , $V(x)$, $M(x)$, $y(x)$ data	~200 data points + embedded SFD and BMD line charts
3 — Material_C omparison	All 10 materials compared	FOS, deflection, weight — Safe/Unsafe icons, alternating rows

Library: openpyxl. Styling: PatternFill, Font, Alignment, Border. Charts: openpyxl.chart.LineChart with Reference objects. All data buffered in BytesIO and served via st.download_button.

15 PDF Report Generator

Uses ReportLab Platypus framework with SimpleDocTemplate. Report contains: Title page, Input Parameters table, Results table, AI Suggestions section, Engineering Diagrams embedded as 120 DPI PNG. Served via st.download_button with MIME type application/pdf.

16 Unit System

Parameter	Display Options	Internal Conversion
Length	mm · cm · m	L_CONV: mm=0.001, cm=0.01, m=1.0 → always stored in metres
Force	N · kN	F_CONV: N=1.0, kN=1000.0 → always stored in Newtons
UDL	N/m · kN/m · N/mm...	$w_{SI} = w_{display} \times F_CONV / L_CONV \rightarrow N/m$
Max Length	100m / 10,000cm / 100,000mm	All internal calculations in SI units always

17 Zoom System — Cross Platform

Platform	Event Used	Method
Android / iOS	touchstart + preventDefault()	Touch event on zoom buttons
Windows / Mac	click event	Mouse click on zoom buttons
Keyboard	keydown listener	Ctrl/Cmd + = (in), – (out), 0 (reset)
Persistence	localStorage	Key 'bsZoom' — restored on page reload
Range	40% to 280%	Step size 15% per click

18 UI/UX — Islands Dark Theme

Inspired by Islands Dark VSCode theme — deep purple/indigo dark background. Google Fonts: Orbitron (headings, futuristic), Rajdhani (body, technical). Gradient text on H1 (blue→purple→teal). Floating metric cards with translateY animation. Neon slider track. Safe box with pulse animation. ANSYS-style stress colormap.

19 Deployment — Streamlit Cloud

Step	Action	Details
1	Push to GitHub	All files including requirements.txt in repo root
2	Go to share.streamlit.io	Login with GitHub account
3	Create New App	Select repo → Branch: main → File: beam_simulator.py
4	Deploy	Streamlit installs packages from requirements.txt automatically
5	App Live	Available at beam-simulator.streamlit.app
Update	Push new code	Streamlit detects changes → go to dashboard → Reboot

Python Version: 3.14.5 | Memory Usage: ~120 MB | Free tier limit: 1 GB

[GitHub: github.com/rupanshujangid-collab/Beam-Simulator](https://github.com/rupanshujangid-collab/Beam-Simulator)

[Live App: beam-simulator.streamlit.app](https://beam-simulator.streamlit.app)

20 Complete Formulas Reference

Reactions — Simply Supported

Parameter	Formula
RA (point loads)	$RA = \sum [P_i \cdot (L - a_i) / L]$
RB (point loads)	$RB = \sum [P_i \cdot a_i / L]$
RA + RB (UDL add-on)	$RA += wL/2 \quad \quad RB += wL/2$

Reactions — Cantilever

Parameter	Formula
RA	$RA = \sum P + w \cdot L$
MA (fixed moment)	$MA = \sum [P \cdot (L-a)] + w \cdot L^2/2$

Reactions — Fixed-Fixed

Parameter	Formula
RA	$RA = \sum [P \cdot (L-a)^2 \cdot (2a+L) / L^3] + wL/2$
RB	$RB = \sum [P \cdot a^2 \cdot (3L-2a) / L^3] + wL/2$
MA	$MA = \sum [P \cdot a \cdot (L-a)^2 / L^2] + wL^2/12$

Section Properties

Parameter	Formula
I — Rectangular	$I = b \cdot h^3 / 12$
I — Circular	$I = \pi \cdot d^4 / 64$
I — I-Beam	$I = bf \cdot ht^3/12 - (bf-tw) \cdot hw^3/12$
■ — T-Beam centroid	$\blacksquare = [Af \cdot (ht-tf/2) + Aw \cdot (hw/2)] / (Af+Aw)$

Stress & Safety

Parameter	Formula
Bending Stress	$\sigma(x,y) = M(x) \cdot y / I$ [Pa]
Max Bending Stress	$\sigma_{\max} = M_{\max} \cdot c / I / 1 \times 10^6$ [MPa]
Factor of Safety	$FOS = \sigma_{\text{yield}} / \sigma_{\max}$
Deflection Serviceability	$y_{\max} < L/200$
Beam Weight	$W = \rho \cdot A_{\text{section}} \cdot L \cdot 9.81$ [N]

21 Known Issues & Fixes

Issue	ModuleNotFoundError: matplotlib
Root Cause	requirements.txt missing from GitHub repo
Fix Applied	Created requirements.txt with all 7 dependencies, pushed to GitHub, rebooted Streamlit

Issue	Content not visible — only background shown
Root Cause	CSS z-index — canvas/div elements overlapping Streamlit content layer
Fix Applied	Removed canvas and floating div; background is pure CSS only now

Issue	use_container_width deprecation warning
Root Cause	Streamlit 1.58 removed this parameter from st.dataframe
Fix Applied	Replaced with width='stretch' parameter

Issue	Zoom not working on Android
Root Cause	mousedown event not triggered on mobile touch screens
Fix Applied	Added touchstart event listener with e.preventDefault() alongside click

Issue	Emoji glyph missing in plot titles
Root Cause	Default matplotlib font does not include emoji Unicode glyphs
Fix Applied	Warning only — no functional impact. Removed emojis from plot titles

Issue	beam_simulator.py read-only on GitHub
Root Cause	Viewing raw file mode instead of edit mode
Fix Applied	Click pencil icon or navigate to /edit/main/beam_simulator.py in URL

Project Complete!

Live App	beam-simulator.streamlit.app
GitHub	github.com/rupanshujangid-collab/Beam-Simulator
Stack	Python · Streamlit · NumPy · Matplotlib · ReportLab · OpenPyXL · Pandas
Features	SFD · BMD · Deflection · Heatmap · 3D · Dynamic · Excel · PDF · AI · Zoom
Validation	All 5 accuracy tests passed
Platforms	Windows · Mac · Android · iOS — fully compatible

[GitHub Repository](#) — [Click to Open](#)

[Live App](#) — [Click to Open](#)